The empirical recurrence for OEIS A186916, proved by transfer matrix

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3 September 2026

Abstract

OEIS A186916 counts arrays over $\{0, \dots, 3\}$ under a condition comparing a statistic of every 2×2 subblock with those of its neighbours, and carries an empirical recurrence of order 68 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The count is a walk count on $S = 11648$ states and is C-finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The conjecture

OEIS A186916 is "Number of (n+1)X3 0..3 arrays with every 2X2 subblock trace equal to exactly one or two horizontal and vertical neighbor 2X2 subblock traces.". It has offset 1 and begins

$$704, 13440, 253312, 4615296, 86388160, 1613838400, 30161413056, 563483938112, \ \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n)=19*a(n-1)-42*a(n-2)+193*a(n-3)+11367*a(n-4)-72730*a(n-5)+643907*a(n-
> 6)-2092699*a(n-7)+2358885*a(n-8)+3857288*a(n-9)-259965739*a(n-10)+923072651*a(n-
> 11)-5781680009*a(n-12)+20491619898*a(n-13)+8894970587*a(n-14)-170675786175*a(n-
> 15)+2368472286552*a(n-16)-8224292736101*a(n-17)+19330903243739*a(n-18)-105830953991576*a(n-
> 19)+57206563938348*a(n-20)+891386946727050*a(n-21)-5108640876870042*a(n-22)+2429363367231296
> 23)-55854110178877757*a(n-24)+119328786504692099*a(n-25)-194223424712175754*a(n-
> 26)-758065151423762303*a(n-27)+4705172061925685671*a(n-28)-19283863911879897678*a(n-
> 29)+58477064112798375671*a(n-30)-115420288738501952295*a(n-31)+226881171676804374579*a(n-
> 32)-227406016994050647282*a(n-33)-758918990518733400415*a(n-34)+3927514331879039801141*a(n-
> 35)-14160407221007454217745*a(n-36)+38834128880307712389668*a(n-37)-84378310829784482421755*a
> 38)+18677799365373777355805*a(n-39)-34241046481815845997659 4*a(n-40)+4827532417101703935096
> 41)-69482122404735638211919 7*a(n-42)+70123760666458670823544 6*a(n-43)-7563631083620843847737 6
> 44)-94170525364037666589782 4*a(n-45)+41948548844399132136147 20*a(n-46)-9313648502362966805603
> 47)+12537306334386180437424128*a(n-48)-22495404254986469017386624*a(n-49)+3422677683993852888
> 50)-24704967584148780932904448*a(n-51)+21360185887183265068596224*a(n-52)-2392630952080660355
> 53)-30199236051497551149019136*a(n-54)+90054299370892913001701376*a(n-55)-1212403379835547571
> 56)+20729557931851413092155 3920*a(n-57)-29402741323012331796155 5968*a(n-58)+3279912412775595
> 59)-36917964439423225469809 4592*a(n-60)+37171637343057824720407 7568*a(n-61)-
> 33302047553278670316817 6128*a(n-62)+30232872488417057346001 3056*a(n-63)-2271552773208938895 61
> 64)+13923945789808084866249 5232*a(n-65)-97256446345939717088870400*a(n-66)+56212147213798847
> 67)-13897379972622097317888000*a(n-68)

As of the "Last modified" line on the live entry (Aug 04 2025 02:13:03, revision 13) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Write σ for the sum of its main diagonal, taken on a 2×2 subblock. The contiguous 2×2 subblocks of an $(n+1) \times 3$ array over $\{0, \ldots, 3\}$ form a grid of n rows and 2 columns, and σ labels that grid. The entry requires that every subblock carries the same value of σ as exactly one or two of its neighbours horizontally or vertically.

3 The count is a walk count

This condition is NOT a condition on a pair. Whether a subblock has a matching neighbour cannot be decided when its own row is placed: the row BELOW it may yet supply the match. So the state carries, besides the 2 consecutive array rows, a tally for each of the 2 subblocks of the row just completed — a count of its matches so far, capped at three, three being already too many — counting its matches among its own row and the row above.

A step appends an array row. It fixes the next subblock row, adds the matches between the two rows to the OLD row's tallies, and requires those totals to be acceptable, the old row being final at that moment; the new row's tallies start from its own horizontal matches plus the matches just made. A state is ACCEPTING when its own tally is already acceptable, which is exactly the right condition for the last subblock row, the one with nothing below it. Only the states whose tally came from their own row may BEGIN an array, since the first row has nothing above it.

An array of $n+1$ rows is then a walk of $n-1$ steps and

$$a(n) \;=\; \iota^{\top} M^{\,n-1} \tau$$

with ι the indicator of the starting states and τ of the accepting ones, on $S = 11648$ states in all. Getting the two vectors right is the whole of the argument: with ι taken as all-ones the first row would be allowed a match from a row above it that does not exist, and the count would be too large. The entry's own first terms settle it — for the smallest cases the model reproduces every published term.

Over the alphabet $\{0, \ldots, 3\}$ the statistic takes values in $[0, 6]$, so the grid it labels carries at most 7 different labels. At $n = 1$ the array has 2 rows and the grid is a single row of 2 subblocks, with only the neighbours inside that row; the entry's first term is 704, which the model reproduces along with every later term, and that is the cheapest check that the grid has been read with the right shape.

In particular a is C-finite. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^{68} - \sum_i c_i t^{68-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^j q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M) \tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+68} - \sum_i c_i M^{j+68-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) \;=\; 19\,a(n{-}1){-}42\,a(n{-}2){+}193\,a(n{-}3){+}11367\,a(n{-}4){-}72730\,a(n{-}5){+}643907\,a(n{-}6){-}2092699\,a(n{-}7){+}$$

for every $n > 68$.

Proof. The vector $w = q(M)\tau$ was formed by 68 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 68 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A186916.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] J. E. Hopcroft, R. Motwani and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*, 3rd ed., Addison–Wesley, 2006.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.